

Comoving Punctures

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I. INTRODUCTION

II. COMOVING COORDINATES, TETRADS, AND LOCAL STF TENSOR MULTIPOLES

Let γ be a timelike worldline with proper time T , written as

$$x^\mu = z^\mu(T). \quad (\text{II.1})$$

We work with mostly-plus signature,

$$g_{\mu\nu}u^\mu u^\nu = -1, \quad \eta_{AB} = \text{diag}(-1, +1, +1, +1), \quad (\text{II.2})$$

and define the four-velocity by

$$u^\mu := \frac{dz^\mu}{dT}. \quad (\text{II.3})$$

A local orthonormal tetrad along the worldline is a set of vectors

$$e_A^\mu(T), \quad A = 0, 1, 2, 3, \quad (\text{II.4})$$

satisfying

$$g_{\mu\nu}e_A^\mu e_B^\nu = \eta_{AB}. \quad (\text{II.5})$$

We choose the timelike tetrad vector to coincide with the four-velocity,

$$e_0^\mu = u^\mu, \quad (\text{II.6})$$

while the spatial triad e_i^μ , $i = 1, 2, 3$, spans the local rest space orthogonal to u^μ :

$$g_{\mu\nu}u^\mu e_i^\nu = 0, \quad g_{\mu\nu}e_i^\mu e_j^\nu = \delta_{ij}. \quad (\text{II.7})$$

The spatial projector orthogonal to u^μ is

$$h_{\mu\nu} := g_{\mu\nu} + u_\mu u_\nu, \quad (\text{II.8})$$

and in the tetrad frame it is represented by

$$h_{\mu\nu} = \delta_{ij}e^i_\mu e^j_\nu. \quad (\text{II.9})$$

A. Comoving local coordinates

Near the worldline one may introduce local spatial coordinates X^i by shooting geodesics orthogonal to γ . Schematically,

$$x^\mu = \exp_{z(T)} (X^i e_i{}^\mu(T)). \quad (\text{II.10})$$

Thus T labels the point on the reference worldline, while X^i label displacement in the instantaneous local rest frame. On the worldline,

$$\left(\frac{\partial}{\partial T}\right)_\gamma^\mu = u^\mu, \quad \left(\frac{\partial}{\partial X^i}\right)_\gamma^\mu = e_i{}^\mu. \quad (\text{II.11})$$

A particularly useful choice is to extend u^μ away from the worldline as a timelike congruence and adapt the time coordinate to this congruence:

$$\boxed{\left(\frac{\partial}{\partial T}\right)^\mu = u^\mu} \quad (\text{II.12})$$

not only on γ , but throughout a neighborhood of the worldline. Equivalently, the local spatial coordinates are Lie-dragged along the flow of u^μ :

$$u^\mu \nabla_\mu T = 1, \quad u^\mu \nabla_\mu X^i = 0. \quad (\text{II.13})$$

Hence the coordinates X^i are comoving labels, while T measures proper time along the congruence.

B. Transport of the tetrad

The spatial triad may be transported along the worldline in several ways. A natural non-rotating choice is Fermi–Walker transport. Define the acceleration

$$a^\mu := u^\nu \nabla_\nu u^\mu. \quad (\text{II.14})$$

With mostly-plus signature, a^μ is spatial:

$$u_\mu a^\mu = 0. \quad (\text{II.15})$$

A Fermi–Walker transported spatial triad satisfies

$$\frac{D e_i{}^\mu}{dT} = u^\mu a_\nu e_i{}^\nu. \quad (\text{II.16})$$

Equivalently, defining the spatial acceleration components

$$a_i := a_\mu e_i^\mu, \quad (II.17)$$

we have

$$\frac{De_i^\mu}{dT} = a_i u^\mu. \quad (II.18)$$

This preserves

$$u_\mu e_i^\mu = 0, \quad e_i^\mu e_{j\mu} = \delta_{ij}. \quad (II.19)$$

More generally, one may allow the spatial triad to rotate with angular velocity Ω^i :

$$\frac{De_i^\mu}{dT} = a_i u^\mu + \epsilon_{ijk} \Omega^j e_k^\mu. \quad (II.20)$$

For a simple local multipole description, one usually chooses a non-rotating or otherwise adapted triad so that the tetrad components of the field have no unnecessary time dependence.

C. Comoving stationarity

Let $F_{\mu_1 \dots \mu_p}$ be a tensor field near the worldline. Its tetrad components are

$$F_{A_1 \dots A_p} := e_{A_1}^{\mu_1} \dots e_{A_p}^{\mu_p} F_{\mu_1 \dots \mu_p}. \quad (II.21)$$

If the field is stationary in the comoving frame, then the invariant statement is

$$\mathcal{L}_u F_{\mu_1 \dots \mu_p} = 0. \quad (II.22)$$

When the time coordinate is adapted so that

$$\left(\frac{\partial}{\partial T} \right)^\mu = u^\mu, \quad (II.23)$$

this condition becomes, in the comoving coordinate system,

$$\partial_T F_{\mu_1 \dots \mu_p} = 0. \quad (II.24)$$

If the tetrad is also chosen to be comoving, in the sense that its rotation has been fixed so that the tetrad components are not artificially time-dependent, then

$$\partial_T F_{A_1 \dots A_p} = 0. \quad (II.25)$$

Thus the field may be described entirely by its spatial dependence in the local rest frame:

$$F_{A_1 \dots A_p} = F_{A_1 \dots A_p}(X^i). \quad (II.26)$$

D. Local spherical basis

Introduce the local Euclidean radius and radial unit vector

$$R := \sqrt{\delta_{ij} X^i X^j}, \quad n^i := \frac{X^i}{R}. \quad (\text{II.27})$$

Then

$$\delta_{ij} n^i n^j = 1. \quad (\text{II.28})$$

The spatial metric on the local rest space is δ_{ij} , so all STF operations are taken with respect to δ_{ij} .

The local spherical basis is

$$\{e_R^i, e_\theta^i, e_\phi^i\}, \quad (\text{II.29})$$

with

$$e_R^i = n^i. \quad (\text{II.30})$$

The angular metric on the unit sphere is

$$\Omega_{AB} d\theta^A d\theta^B = d\theta^2 + \sin^2 \theta d\phi^2, \quad (\text{II.31})$$

where $A, B = \theta, \phi$. The unit radial vector $n^i(\theta, \phi)$ maps tensors in the local Cartesian rest frame to angular tensors on S^2 .

E. Why the condition $\partial_T = u$ is useful

Suppose a scalar field Φ is stationary in the comoving frame. The invariant statement is

$$\mathcal{L}_u \Phi = u^\mu \nabla_\mu \Phi = 0. \quad (\text{II.32})$$

If the time coordinate is chosen so that

$$\left(\frac{\partial}{\partial T} \right)^\mu = u^\mu, \quad (\text{II.33})$$

then this becomes simply

$$\partial_T \Phi = 0. \quad (\text{II.34})$$

Therefore the field has no explicit dependence on the comoving time coordinate:

$$\Phi = \Phi(X^i). \quad (\text{II.35})$$

This is the main simplification. The field is described on a single local spatial slice and then Lie-dragged along the comoving flow.

For tensor fields, one first projects onto the tetrad:

$$F_{A_1 \dots A_p} := e_{A_1}^{\mu_1} \dots e_{A_p}^{\mu_p} F_{\mu_1 \dots \mu_p}. \quad (\text{II.36})$$

If both the field and the tetrad basis are comoving in the appropriate sense, then the tetrad components satisfy

$$\partial_T F_{A_1 \dots A_p} = 0. \quad (\text{II.37})$$

Thus the local tensor components may also be treated as functions only of the spatial coordinates X^i .

F. Scalar STF multipoles

For a scalar field independent of T , one may write

$$\Phi = \Phi(R, n^i). \quad (\text{II.38})$$

Using the multi-index notation

$$L = i_1 \dots i_\ell, \quad n^L = n^{i_1} \dots n^{i_\ell}, \quad (\text{II.39})$$

and denoting STF projection by angle brackets,

$$n^{\langle L \rangle} := n^{\langle i_1 \dots i_\ell \rangle}, \quad (\text{II.40})$$

the scalar multipole expansion is

$$\boxed{\Phi(R, n^i) = \sum_{\ell=0}^{\infty} \Phi_L(R) n^{\langle L \rangle}} \quad (\text{II.41})$$

where

$$\Phi_L = \Phi_{\langle i_1 \dots i_\ell \rangle}. \quad (\text{II.42})$$

This is equivalent to the usual spherical harmonic expansion

$$\Phi(R, \theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \Phi_{\ell m}(R) Y_{\ell m}(\theta, \phi). \quad (\text{II.43})$$

G. Tensor-valued STF multipoles

For a spatial tensor field, for example a vector V_i , a symmetric tensor S_{ij} , or the spatial components of a spacetime tensor projected onto the tetrad, one expands the tensor components in angular STF tensors.

For a spatial vector,

$$V_i = V_i(R, n^j), \quad (\text{II.44})$$

one may write the STF expansion

$$V_i(R, n^j) = \sum_{\ell=0}^{\infty} V_{iL}(R) n^{\langle L \rangle} \quad (\text{II.45})$$

where the coefficients V_{iL} are STF in the multipole indices L :

$$V_{iL} = V_{i\langle L \rangle}. \quad (\text{II.46})$$

The index i is the tensor index of the field, while L labels the angular multipole structure.

Similarly, for a spatial rank-two tensor,

$$S_{ij} = S_{ij}(R, n^k), \quad (\text{II.47})$$

one writes

$$S_{ij}(R, n^k) = \sum_{\ell=0}^{\infty} S_{ijL}(R) n^{\langle L \rangle} \quad (\text{II.48})$$

with

$$S_{ijL} = S_{ij\langle L \rangle}. \quad (\text{II.49})$$

If S_{ij} is symmetric, tracefree, or transverse, these properties impose corresponding algebraic or differential constraints on the coefficients S_{ijL} .

For a general tetrad-projected tensor field,

$$F_{A_1 \dots A_p}(R, n^i), \quad (\text{II.50})$$

the local STF expansion takes the form

$$F_{A_1 \dots A_p}(R, n^i) = \sum_{\ell=0}^{\infty} F_{A_1 \dots A_p L}(R) n^{\langle L \rangle} \quad (\text{II.51})$$

where

$$F_{A_1 \dots A_p L} = F_{A_1 \dots A_p \langle L \rangle}. \quad (\text{II.52})$$

Thus the tensorial nature of the field is carried by the tetrad indices

$$A_1, \dots, A_p, \quad (\text{II.53})$$

while the angular dependence is carried by the STF multipole index L .

For $s \neq 0$ scalars ${}_s\Phi$ we use the helicity basis

$$\hat{m}_i^\pm = \frac{1}{\sqrt{2}}(\hat{\theta}_i \pm i\hat{\varphi}_i) \quad (\text{II.54})$$

and create spin $s = \pm 1$ weighted building blocks via

$$\hat{m}_i^+ \hat{n}^{\langle iL-1 \rangle}, \quad \hat{m}_i^- \hat{n}^{\langle iL-1 \rangle} \quad (\text{II.55})$$

and the rule

$${}_s\Phi(R, n^i) = \sum_{\ell \geq |s|} \Phi_L^{(s)}(R) m_{i_1}^{\sigma_1} \dots m_{i_s}^{\sigma_s} \hat{n}^{\langle i_1 \dots i_s L-s \rangle} \quad (\text{II.56})$$

with $\sigma_k = \text{sgn}(s)$.

Using the

H. Connection with tensor spherical harmonics

The STF expansion is the Cartesian version of an expansion in tensor spherical harmonics. For example, a vector field on the local rest space may be decomposed into radial, electric-parity, and magnetic-parity vector harmonics:

$$V_i(R, \theta, \phi) = \sum_{\ell m} [V_{\ell m}^R(R) Y_{\ell m} n_i + V_{\ell m}^E(R) D_i Y_{\ell m} + V_{\ell m}^B(R) \epsilon_i^j D_j Y_{\ell m}] . \quad (\text{II.57})$$

Here D_i denotes the covariant derivative on the unit sphere after projecting into the angular directions, and ϵ_{ij} is the volume form on S^2 .

For a symmetric spatial rank-two tensor, one obtains the tensor-harmonic decomposition

$$\begin{aligned} S_{ij} = \sum_{\ell m} & \left[S_{\ell m}^{RR}(R) Y_{\ell m} n_i n_j + S_{\ell m}^{RE}(R) n_{(i} D_{j)} Y_{\ell m} + S_{\ell m}^{RB}(R) n_{(i} \epsilon_{j)}^k D_k Y_{\ell m} \right. \\ & + S_{\ell m}^T(R) \Omega_{ij} Y_{\ell m} + S_{\ell m}^E(R) \left(D_i D_j Y_{\ell m} + \frac{1}{2} \ell(\ell+1) \Omega_{ij} Y_{\ell m} \right) \\ & \left. + S_{\ell m}^B(R) \epsilon_{(i}^k D_{j)} D_k Y_{\ell m} \right]. \end{aligned} \quad (\text{II.58})$$

See the appendix ?? for details on the spin-weighted case.

I. Local Taylor expansion in STF form

For a smooth scalar field near the worldline,

$$\Phi(X^i) = \sum_{\ell=0}^{\infty} \frac{1}{\ell!} \Phi_L X^{\langle L \rangle}, \quad (\text{II.59})$$

where

$$\Phi_L = \partial_{\langle L \rangle} \Phi \big|_{X^i=0}. \quad (\text{II.60})$$

For a tensor field,

$$F_{A_1 \dots A_p}(X^i) = \sum_{\ell=0}^{\infty} \frac{1}{\ell!} F_{A_1 \dots A_p L} X^{\langle L \rangle}, \quad (\text{II.61})$$

with

$$F_{A_1 \dots A_p L} = \partial_{\langle L \rangle} F_{A_1 \dots A_p} \big|_{X^i=0}. \quad (\text{II.62})$$

This gives the local STF multipole moments of the tensor field in the comoving tetrad frame.

Assuming comoving stationarity, the first order singular field has the general expansion

$$\Phi(R, n^i) = \frac{q}{R} + \sum_{n \geq 0} R^n \sum_{\ell=0}^{\ell_{\max}(n)} \mathcal{Q}_L n^{\langle L \rangle}, \quad (\text{II.63})$$

where $\ell_{\max}(n) = 3n + 3$ while singular vector and tensor metric components obey similar relations. It is precisely this homogenous scaling in R at each puncture order which simplifies the subsequent analysis.

III. NOTEBOOK IMPLEMENTATION OF THE CO-MOVING COORDINATES

The mystery m -mode notebook (from Patrick but origin unknown) introduces two coordinate charts on the same four-dimensional manifold M . The first is the Boyer–Lindquist chart,

$$x_{\text{BL}}^\mu = (t, r, \theta, \phi),$$

and the second is the co-moving chart,

$$X_{\text{CM}}^A = (T, X, Y, Z).$$

The goal is to implement the local coordinate system used in the puncture construction, where the particle sits at the origin of the spatial coordinates,

$$X = Y = Z = 0.$$

This is the coordinate system used to express the tetrad components of the singular puncture field in the universal local form

$$h_{(a)(b)}^P = \sum_{i,j,k,p,q} c_{ijkpq}^{(a)(b)} X^i Y^j Z^k R^p \log^q R, \quad R^2 = X^2 + Y^2 + Z^2. \quad (\text{III.1})$$

A. Orbit constants and dimensionless variables

The notebook defines constants

$$M, \quad r_0, \quad a, \quad m, \quad \lambda,$$

together with orbit-evaluated quantities

$$\mathcal{B}, \quad \mathcal{D}, \quad z_c, \quad \Omega, \quad \alpha, \quad \rho_0.$$

The dimensionless variables are introduced by

$$r_0 = M\rho_0, \quad a = \alpha\sqrt{Mr_0} = M\alpha\sqrt{\rho_0}.$$

Thus ρ_0 is the orbital radius in units of M , while α is the dimensionless spin parameter used by the notebook.

The notebook defines

$$\mathcal{D} = \sqrt{\frac{2\alpha + \rho_0 - 3}{\rho_0}},$$

$$\mathcal{B} = \sqrt{\frac{\alpha^2 + \rho_0 - 2}{\rho_0}},$$

and

$$\Omega = \frac{1}{M\sqrt{\rho_0}(\alpha + \rho_0)}.$$

Equivalently, before replacing $r_0 = M\rho_0$ and $a = \alpha\sqrt{Mr_0}$,

$$\Omega = \frac{\sqrt{M}}{a\sqrt{M} + r_0^{3/2}}. \quad (\text{III.2})$$

The length scale z_c appearing in the azimuthal co-moving coordinate is implemented as

$$z_c = 2 \frac{\sqrt{M}}{\sqrt{r_0}} \frac{\mathcal{B}}{\Omega \mathcal{D}}.$$

After using $r_0 = M\rho_0$, this becomes

$$z_c = \frac{2\mathcal{B}}{\sqrt{\rho_0}\Omega\mathcal{D}} = 2M(\alpha + \rho_0) \frac{\mathcal{B}}{\mathcal{D}}.$$

The Schwarzschild limit used in the notebook is

$$\alpha \rightarrow 0, \quad a \rightarrow 0. \quad (\text{III.3})$$

B. The co-moving phase

The notebook uses the co-rotating phase

$$\psi := \phi - \Omega t. \quad (\text{III.4})$$

In terms of the original Boyer–Lindquist variables,

$$\psi = \phi - \frac{\sqrt{M}}{a\sqrt{M} + r_0^{3/2}} t.$$

The particle is located at

$$r = r_0, \quad \theta = \frac{\pi}{2}, \quad \psi = 0.$$

Therefore

$$X = 0, \quad Y = 0, \quad Z = 0$$

is precisely the particle limit.

C. Coordinate transformation

The notebook defines the co-moving spatial coordinates by

$$X = \frac{r - r_0}{\mathcal{B}}, \quad Y = -r_0 \cos \theta, \quad (\text{III.5a})$$

and

$$Z = z_c \sin\left(\frac{\psi}{2}\right) = z_c \sin\left[\frac{1}{2} \left(\phi - \frac{\sqrt{M}}{a\sqrt{M} + r_0^{3/2}} t \right)\right]. \quad (\text{III.5b})$$

The time coordinate is a linear combination of t and ϕ . Define

$$\Delta_T := 2a\sqrt{M} + \sqrt{r_0}(r_0 - 3M),$$

$$A_t := a\sqrt{M} + \sqrt{r_0}(r_0 - 2M),$$

and

$$A_\phi := \sqrt{M} \left(a^2 + r_0^2 - 2a\sqrt{Mr_0} \right).$$

Then

$$T = \mathcal{D} \frac{A_t t - A_\phi \phi}{\Delta_T}.$$

Explicitly,

$$T = \mathcal{D} \frac{\left(a\sqrt{M} + \sqrt{r_0}(r_0 - 2M)\right) t - \sqrt{M} (a^2 + r_0^2 - 2a\sqrt{Mr_0}) \phi}{2a\sqrt{M} + \sqrt{r_0}(r_0 - 3M)}. \quad (\text{III.6})$$

Thus the full map is

$$(t, r, \theta, \phi) \mapsto (T, X, Y, Z),$$

with

$$\boxed{\begin{aligned} T &= \mathcal{D} \frac{\left(a\sqrt{M} + \sqrt{r_0}(r_0 - 2M)\right) t - \sqrt{M} (a^2 + r_0^2 - 2a\sqrt{Mr_0}) \phi}{2a\sqrt{M} + \sqrt{r_0}(r_0 - 3M)}, \\ X &= \frac{r - r_0}{\mathcal{B}}, \\ Y &= -r_0 \cos \theta, \\ Z &= z_c \sin \left[\frac{1}{2} (\phi - \Omega t) \right]. \end{aligned}}$$

D. Local distance variables

The local Euclidean distance used in the puncture expansion is

$$R_{\text{loc}}^2 = X^2 + Y^2 + Z^2.$$

Since

$$Z = z_c \sin \left(\frac{\psi}{2} \right),$$

we have

$$R_{\text{loc}}^2 = X^2 + Y^2 + z_c^2 \sin^2 \left(\frac{\psi}{2} \right).$$

It is useful to define

$$\rho^2 := X^2 + Y^2. \quad (\text{III.7})$$

Then

$$R_{\text{loc}}^2 = \rho^2 + z_c^2 \sin^2 \left(\frac{\psi}{2} \right).$$

For the m -mode integral formulas, the notebook also introduces

$$\mathcal{R} := \sqrt{\rho^2 + z_c^2}. \quad (\text{III.8})$$

This is not the same object as R_{loc} . Rather, $\mathcal{R}$ is the second endpoint radius appearing after rewriting the angular dependence in the form

$$R_{\text{loc}}^2 = a - b \cos \psi,$$

with

$$a = \rho^2 + \frac{z_c^2}{2} = \frac{\mathcal{R}^2 + \rho^2}{2}, \quad b = \frac{z_c^2}{2} = \frac{\mathcal{R}^2 - \rho^2}{2}.$$

The Legendre argument used in the analytic m -mode integrals is

$$z = \frac{a}{\sqrt{a^2 - b^2}} = \frac{\rho^2 + \frac{1}{2}z_c^2}{\rho\sqrt{\rho^2 + z_c^2}} = \frac{\mathcal{R}^2 + \rho^2}{2\mathcal{R}\rho}.$$

This is precisely the variable implemented in the notebook as

$$z \mapsto \frac{X^2 + Y^2 + \frac{1}{2}z_c^2}{\sqrt{X^2 + Y^2}\sqrt{X^2 + Y^2 + z_c^2}}.$$

E. The Z' variable

We replace higher powers of Z by powers of $R^2 - X^2 - Y^2$, leaving at most one explicit odd power of Z . The notebook then introduces the single-valued angular variable

$$Z' := z_c \sin \psi. \tag{III.9}$$

Since

$$Z = z_c \sin \left(\frac{\psi}{2} \right), \tag{III.10}$$

we have the local relation

$$Z = \frac{Z'}{2} + \frac{(Z')^3}{16z_c^2} + \dots$$

Therefore, after truncating to the desired puncture order, the puncture field can be rewritten in the reduced form

$$h^{P\mu\nu} = \sum_{i,j,k,p,q} d_{ijkpq}^{\mu\nu} X^i Y^j (Z')^k R_{\text{loc}}^p \log^q R_{\text{loc}}, \quad k, q \in \{0, 1\}. \tag{III.11}$$

The notebook implements this by using both

$$Z = z_c \sin \left(\frac{\psi}{2} \right)$$

and

$$Z' = z_c \sin \psi.$$

F. Jacobian of the chart transformation

The Jacobian between the Boyer–Lindquist and co-moving charts. Let

$$J_\mu{}^A := \frac{\partial X_{\text{CM}}^A}{\partial x_{\text{BL}}^\mu},$$

where

$$x_{\text{BL}}^\mu = (t, r, \theta, \phi), \quad X_{\text{CM}}^A = (T, X, Y, Z).$$

The entries are

$$J_\mu{}^A = \begin{pmatrix} \mathcal{D} \frac{a\sqrt{M} + \sqrt{r_0}(r_0 - 2M)}{2a\sqrt{M} + \sqrt{r_0}(r_0 - 3M)} & 0 & 0 & -\frac{\sqrt{M} z_c}{2(a\sqrt{M} + r_0^{3/2})} \cos\left(\frac{\psi}{2}\right) \\ 0 & \mathcal{B}^{-1} & 0 & 0 \\ 0 & 0 & r_0 \sin \theta & 0 \\ -\mathcal{D} \frac{\sqrt{M}(a^2 + r_0^2 - 2a\sqrt{M}r_0)}{2a\sqrt{M} + \sqrt{r_0}(r_0 - 3M)} & 0 & 0 & \frac{z_c}{2} \cos\left(\frac{\psi}{2}\right) \end{pmatrix}. \quad (\text{III.12})$$

For a covariant tensor written in the co-moving chart,

$$h_{AB}^{\text{CM}},$$

the corresponding Boyer–Lindquist components are obtained by pullback:

$$h_{\mu\nu}^{\text{BL}} = J_\mu{}^A J_\nu{}^B h_{AB}^{\text{CM}}. \quad (\text{III.13})$$

Similarly, for derivatives,

$$\partial_\mu^{\text{BL}} = J_\mu{}^A \partial_A^{\text{CM}}.$$

IV. LOCAL PUNCTURE INTEGRALS AND THEIR m -MODE REDUCTION

The puncture field is first written in a coordinate system co-moving with the particle,

$$x^\mu = (T, X, Y, Z),$$

where, schematically,

$$X = \frac{\Delta r}{\mathcal{B}}, \quad Y \sim -r_p \cos \theta, \quad Z = z_c \sin\left(\frac{\phi - \Omega t}{2}\right).$$

It is also useful to introduce

$$\rho^2 := X^2 + Y^2, \quad R^2 := X^2 + Y^2 + Z^2 = \rho^2 + Z^2.$$

The tensor components are projected onto the Kinnersley tetrad,

$$e_{(a)}^\mu = (l^\mu, n^\mu, m^\mu, \bar{m}^\mu), \quad h_{(a)(b)}^P := e_{(a)}^\mu e_{(b)}^\nu h_{\mu\nu}^P.$$

In these variables the local singular/puncture field has the universal monomial structure

$$h_{(a)(b)}^P = \sum_{i,j,k,p,q} c_{ijkpq}^{(a)(b)} X^i Y^j Z^k R^p \log^q R.$$

The detailed geometry is hidden in the coefficients c_{ijkpq}^{ab} . The nontrivial part of the m -mode decomposition is therefore reduced to scalar integrals of monomials of the form

$$X^i Y^j Z^k R^p \log^q R.$$

For the different puncture pieces the powers have the following structure:

$$h_{(a)(b)}^{SS} : \quad p \text{ even}, \quad q \in \{0, 1\},$$

whereas

$$h_{(a)(b)}^{P1} \text{ and } h_{(a)(b)}^{SR} : \quad p \text{ odd}, \quad q = 0.$$

To make the mode decomposition simpler, higher powers of Z are traded for powers of $R^2 - X^2 - Y^2$. Since

$$Z^2 = R^2 - X^2 - Y^2,$$

one uses

$$Z^n = Z (R^2 - X^2 - Y^2)^{(n-1)/2}, \quad n \text{ odd}, \quad n > 1,$$

and

$$Z^n = (R^2 - X^2 - Y^2)^{n/2}, \quad n \text{ even}, \quad n > 0.$$

Thus all higher powers of Z can be absorbed into new coefficients and new powers of X, Y, R , leaving at most one explicit odd power of Z .

Next introduce the single-valued angular variable

$$Z' := z_c \sin(\phi - \Omega t).$$

Near the particle,

$$Z = \frac{Z'}{2} + \frac{(Z')^3}{16} + \dots$$

After re-expanding and truncating to the required puncture order, the field can be written in the reduced form

$$h_{(a)(b)}^P = \sum_{i,j,k,p,q} d_{ijkpq}^{(a)(b)} X^i Y^j (Z')^k R^p \log^q R, \quad k, q \in \{0, 1\}.$$

Hence only the cases with no Z' , one Z' , no logarithm, or one logarithm are needed.

The m -mode is defined by

$$h_{(a)(b)}^{P,m} = \frac{1}{2\pi} \int_{-\pi}^{\pi} h_{(a)(b)}^P e^{-im\phi} d\phi.$$

Substituting the reduced monomial form gives

$$h_{(a)(b)}^{P,m} = \frac{1}{2\pi} \sum_{i,j,k,p,q} d_{ijkpq}^{(a)(b)} X^i Y^j \int_{-\pi}^{\pi} (Z')^k R^p \log^q R e^{-im\phi} d\phi.$$

After shifting to the co-moving angle

$$\psi := \phi - \Omega t,$$

one has

$$e^{-im\phi} = e^{-im\Omega t} e^{-im\psi}.$$

The overall phase $e^{-im\Omega t}$ can be pulled out, so the universal scalar integrals are

$$\mathcal{I}_{p,m}^{(k,q)} := \int_{-\pi}^{\pi} (z_c \sin \psi)^k R(\psi)^p \log^q R(\psi) e^{-im\psi} d\psi, \quad k, q \in \{0, 1\}.$$

Thus the full mode decomposition reduces to four families:

$$\begin{aligned} I_{p,m} &:= \int_{-\pi}^{\pi} R^p e^{-im\psi} d\psi, \\ S_{p,m} &:= \int_{-\pi}^{\pi} R^p \sin \psi e^{-im\psi} d\psi, \\ L_m &:= \int_{-\pi}^{\pi} \log R e^{-im\psi} d\psi, \end{aligned}$$

and

$$M_m := \int_{-\pi}^{\pi} \log R \sin \psi e^{-im\psi} d\psi.$$

The sine insertion is not really independent. Since

$$\sin \psi e^{-im\psi} = \frac{1}{2i} (e^{-i(m-1)\psi} - e^{-i(m+1)\psi}),$$

we have the exact shift relations

$$S_{p,m} = \frac{1}{2i} (I_{p,m-1} - I_{p,m+1}),$$

and

$$M_m = \frac{1}{2i} (L_{m-1} - L_{m+1}).$$

So the four families can already be reduced to the two master families

$$I_{p,m}, \quad L_m.$$

For the algebraic family one obtains, up to normalization conventions,

$$I_{p,m} \sim \rho^{p/2} \mathcal{R}^{p/2} P_{p/2}^{|m|}(z),$$

where

$$\rho := \sqrt{X^2 + Y^2}, \quad \mathcal{R} := \sqrt{X^2 + Y^2 + z_c^2},$$

and $z = z(\mathcal{R}, \rho, z_c)$ is the corresponding dimensionless Legendre argument. In the standard law-of-cosines form

$$R(\psi)^2 = \mathcal{R}^2 + \rho^2 - 2\mathcal{R}\rho \cos \psi,$$

one has

$$z = \frac{\mathcal{R}^2 + \rho^2}{2\mathcal{R}\rho}.$$

The logarithmic family can be obtained by differentiating the algebraic family with respect to the exponent:

$$\frac{\partial}{\partial p} R^p = R^p \log R.$$

Hence

$$L_m = \left. \frac{\partial}{\partial p} I_{p,m} \right|_{p=0}.$$

This is a very useful way to avoid treating the logarithmic integrals as a separate problem.

Schematically, the result has the form

$$L_m \sim \delta_{m0} \log(\mathcal{R} - \rho) + \delta_{|m| \geq 1} (\mathcal{R} + \rho)^{-2|m|},$$

up to normalization and convention-dependent factors. The logarithmic sine integral then follows from the same shift relation,

$$M_m = \frac{1}{2i} (L_{m-1} - L_{m+1}).$$

The main lesson is that the tensorial puncture problem reduces to a finite linear combination of scalar Fourier coefficients:

$$h_{ab}^{P,m} = \sum_{i,j,p} [A_{ijp}^{ab} X^i Y^j I_{p,m} + B_{ijp}^{ab} X^i Y^j I_{p,m\pm 1} + C_{ij}^{ab} X^i Y^j L_m + D_{ij}^{ab} X^i Y^j L_{m\pm 1}], \quad (\text{IV.1})$$

where the shifted modes arise from the single allowed power of $Z' \sim \sin \psi$.

V. PRECISE FORM OF THE PUNCTURE m -MODE INTEGRALS

Let us introduce the angle-dependent local distance

$$r(\psi)^2 := X^2 + Y^2 + z_c^2 \sin^2\left(\frac{\psi}{2}\right) = \varrho^2 + z_c^2 \sin^2\left(\frac{\psi}{2}\right).$$

Equivalently,

$$r(\psi)^2 = a - b \cos \psi,$$

where

$$a := \varrho^2 + \frac{z_c^2}{2} = \frac{\mathcal{R}^2 + \varrho^2}{2}, \quad b := \frac{z_c^2}{2} = \frac{\mathcal{R}^2 - \varrho^2}{2}.$$

We shall also use

$$D := \sqrt{a^2 - b^2} = \mathcal{R}\varrho,$$

and

$$z := \frac{a}{D} = \frac{\mathcal{R}^2 + \varrho^2}{2\mathcal{R}\varrho}.$$

Notice that $z > 1$ away from $\varrho = 0$. This is the Legendre argument that will appear in the mode integrals.

A. Local monomial structure

Recall after projecting onto the Kinnersley tetrad,

$$e_a^\mu = (l^\mu, n^\mu, m^\mu, \bar{m}^\mu), \quad h_{ab}^P := e_a^\mu e_b^\nu h_{\mu\nu}^P,$$

and expanding

$$Z = z_c \sin\left(\frac{\psi}{2}\right) = \frac{Z'}{2} + \frac{(Z')^3}{16z_c^2} + \cdots,$$

the puncture field has local form

$$h_{(a)(b)}^P = \sum_{i,j,k,p,q} d_{ijkpq}^{(a)(b)} X^i Y^j (Z')^k r^p \log^q r, \quad k, q \in \{0, 1\}.$$

B. Definition of the m -modes

The m -mode is

$$h_{(a)(b)}^{P,m} = \frac{1}{2\pi} \int_{-\pi}^{\pi} h_{(a)(b)}^P e^{-im\phi} d\phi. \quad (\text{V.1})$$

Using $\phi = \psi + \Omega t$, this becomes

$$h_{(a)(b)}^{P,m} = e^{-im\Omega t} \frac{1}{2\pi} \int_{-\pi}^{\pi} h_{(a)(b)}^P(\psi) e^{-im\psi} d\psi. \quad (\text{V.2})$$

Substituting the reduced monomial expansion gives

$$h_{(a)(b)}^{P,m} = e^{-im\Omega t} \sum_{i,j,k,p,q} d_{ijkpq}^{(a)(b)} X^i Y^j \mathcal{I}_{p,m}^{(k,q)}, \quad (\text{V.3})$$

where

$$\mathcal{I}_{p,m}^{(k,q)} := \frac{1}{2\pi} \int_{-\pi}^{\pi} (z_c \sin \psi)^k r(\psi)^p \log^q r(\psi) e^{-im\psi} d\psi. \quad (\text{V.4})$$

Because $k, q \in \{0, 1\}$, there are four basic types:

$$\mathcal{I}_{p,m}^{(0,0)}, \quad \mathcal{I}_{p,m}^{(1,0)}, \quad \mathcal{I}_{p,m}^{(0,1)}, \quad \mathcal{I}_{p,m}^{(1,1)}. \quad (\text{V.5})$$

C. Master algebraic integral

Define

$$J_{\nu m}(a, b) := \frac{1}{2\pi} \int_{-\pi}^{\pi} (a - b \cos \psi)^\nu e^{-im\psi} d\psi, \quad (\text{V.6})$$

where

$$\nu := \frac{p}{2}. \quad (\text{V.7})$$

Since

$$r(\psi)^p = (a - b \cos \psi)^{p/2}, \quad (\text{V.8})$$

we have

$$\mathcal{I}_{p,m}^{(0,0)} = J_{p/2,m}(a, b). \quad (\text{V.9})$$

The exact associated-Legendre representation is

$$\boxed{J_{\nu m}(a, b) = D^\nu \frac{\Gamma(\nu + 1)}{\Gamma(\nu + |m| + 1)} P_\nu^{|m|}(z)} \quad (\text{V.10})$$

with

$$D = \sqrt{a^2 - b^2} = \mathcal{R}_\varrho, \quad z = \frac{a}{D} = \frac{\mathcal{R}^2 + \varrho^2}{2\mathcal{R}_\varrho}. \quad (\text{V.11})$$

Therefore

$$\boxed{\mathcal{I}_{p,m}^{(0,0)} = (\mathcal{R}_\varrho)^{p/2} \frac{\Gamma(\frac{p}{2} + 1)}{\Gamma(\frac{p}{2} + |m| + 1)} P_{p/2}^{|m|}\left(\frac{\mathcal{R}^2 + \varrho^2}{2\mathcal{R}_\varrho}\right)} \quad (\text{V.12})$$

Equivalently, without the $1/(2\pi)$ normalization,

$$\int_{-\pi}^{\pi} r(\psi)^p e^{-im\psi} d\psi = 2\pi (\mathcal{R}_\varrho)^{p/2} \frac{\Gamma(\frac{p}{2} + 1)}{\Gamma(\frac{p}{2} + |m| + 1)} P_{p/2}^{|m|}\left(\frac{\mathcal{R}^2 + \varrho^2}{2\mathcal{R}_\varrho}\right). \quad (\text{V.13})$$

D. Integral with one factor of Z'

Since

$$Z' = z_c \sin \psi, \quad (\text{V.14})$$

we need

$$\mathcal{I}_{p,m}^{(1,0)} = \frac{z_c}{2\pi} \int_{-\pi}^{\pi} \sin \psi r(\psi)^p e^{-im\psi} d\psi. \quad (\text{V.15})$$

Using

$$\sin \psi e^{-im\psi} = \frac{1}{2i} (e^{-i(m-1)\psi} - e^{-i(m+1)\psi}), \quad (\text{V.16})$$

we get the exact shift identity

$$\boxed{\mathcal{I}_{p,m}^{(1,0)} = \frac{z_c}{2i} [J_{p/2,m-1}(a, b) - J_{p/2,m+1}(a, b)]} \quad (\text{V.17})$$

Substituting the Legendre form gives

$$\boxed{\mathcal{I}_{p,m}^{(1,0)} = \frac{z_c}{2i} (\mathcal{R}_\varrho)^{p/2} \Gamma\left(\frac{p}{2} + 1\right) \left[\frac{P_{p/2}^{|m-1|}(z)}{\Gamma(\frac{p}{2} + |m-1| + 1)} - \frac{P_{p/2}^{|m+1|}(z)}{\Gamma(\frac{p}{2} + |m+1| + 1)} \right]}, \quad (\text{V.18})$$

where

$$z = \frac{\mathcal{R}^2 + \varrho^2}{2\mathcal{R}_\varrho}. \quad (\text{V.19})$$

E. Pure logarithmic integral

Define

$$\Lambda_m := \frac{1}{2\pi} \int_{-\pi}^{\pi} \log r(\psi) e^{-im\psi} d\psi. \quad (\text{V.20})$$

Since

$$\frac{\partial}{\partial p} r(\psi)^p = r(\psi)^p \log r(\psi), \quad (\text{V.21})$$

we have

$$\Lambda_m = \left. \frac{\partial}{\partial p} J_{p/2,m}(a, b) \right|_{p=0}. \quad (\text{V.22})$$

Equivalently,

$$\Lambda_m = \frac{1}{2} \left. \frac{\partial}{\partial \nu} J_{\nu m}(a, b) \right|_{\nu=0}. \quad (\text{V.23})$$

There is also a closed Fourier form. Let

$$\lambda := \frac{b}{a+D} = \frac{\mathcal{R} - \varrho}{\mathcal{R} + \varrho}, \quad 0 < \lambda < 1. \quad (\text{V.24})$$

Then

$$a - b \cos \psi = A (1 - 2\lambda \cos \psi + \lambda^2), \quad (\text{V.25})$$

where

$$A := \frac{a+D}{2} = \frac{(\mathcal{R} + \varrho)^2}{4}. \quad (\text{V.26})$$

Thus

$$\log r(\psi) = \frac{1}{2} \log A + \frac{1}{2} \log (1 - 2\lambda \cos \psi + \lambda^2). \quad (\text{V.27})$$

Using

$$\log (1 - 2\lambda \cos \psi + \lambda^2) = -2 \sum_{n=1}^{\infty} \frac{\lambda^n}{n} \cos(n\psi), \quad (\text{V.28})$$

we get

$$\Lambda_0 = \log \left(\frac{\mathcal{R} + \varrho}{2} \right), \quad (\text{V.29})$$

and, for $|m| \geq 1$,

$$\Lambda_m = -\frac{1}{2|m|} \left(\frac{\mathcal{R} - \varrho}{\mathcal{R} + \varrho} \right)^{|m|}. \quad (\text{V.30})$$

Therefore

$$\mathcal{I}_{0,m}^{(0,1)} = \Lambda_m. \quad (\text{V.31})$$

More generally, if one needs $r^p \log r$, then

$$\mathcal{I}_{p,m}^{(0,1)} = \frac{\partial}{\partial p} J_{p/2,m}(a, b). \quad (\text{V.32})$$

F. Logarithmic integral with one factor of Z'

Finally,

$$\mathcal{I}_{p,m}^{(1,1)} = \frac{z_c}{2\pi} \int_{-\pi}^{\pi} \sin \psi r(\psi)^p \log r(\psi) e^{-im\psi} d\psi. \quad (\text{V.33})$$

The same shift identity gives

$$\boxed{\mathcal{I}_{p,m}^{(1,1)} = \frac{z_c}{2i} [\mathcal{I}_{p,m-1}^{(0,1)} - \mathcal{I}_{p,m+1}^{(0,1)}]}. \quad (\text{V.34})$$

In the pure logarithmic case $p = 0$,

$$\boxed{\mathcal{I}_{0,m}^{(1,1)} = \frac{z_c}{2i} [\Lambda_{m-1} - \Lambda_{m+1}]}. \quad (\text{V.35})$$

Explicitly,

$$\mathcal{I}_{0,0}^{(1,1)} = 0. \quad (\text{V.36})$$

For $m = 1$,

$$\mathcal{I}_{0,1}^{(1,1)} = \frac{z_c}{2i} [\Lambda_0 - \Lambda_2] = \frac{z_c}{2i} \left[\log \left(\frac{\mathcal{R} + \varrho}{2} \right) + \frac{1}{4} \left(\frac{\mathcal{R} - \varrho}{\mathcal{R} + \varrho} \right)^2 \right]. \quad (\text{V.37})$$

For $m \geq 2$,

$$\mathcal{I}_{0,m}^{(1,1)} = \frac{iz_c}{4} \left[\frac{1}{m-1} \left(\frac{\mathcal{R} - \varrho}{\mathcal{R} + \varrho} \right)^{m-1} - \frac{1}{m+1} \left(\frac{\mathcal{R} - \varrho}{\varrho + \mathcal{R}} \right)^{m+1} \right]. \quad (\text{V.38})$$

The negative modes follow from

$$\mathcal{I}_{0,-m}^{(1,1)} = -\mathcal{I}_{0,m}^{(1,1)}. \quad (\text{V.39})$$

G. Final reconstruction formula

The full puncture mode can therefore be written as

$$\begin{aligned} h_{ab}^{P,m} = e^{-im\Omega t} \sum_{i,j,p} X^i Y^j & \left[A_{ijp}^{ab} J_{p/2,m} + B_{ijp}^{ab} \frac{z_c}{2i} (J_{p/2,m-1} - J_{p/2,m+1}) + C_{ijp}^{ab} \partial_p J_{p/2,m} \right. \\ & \left. + D_{ijp}^{ab} \frac{z_c}{2i} (\partial_p J_{p/2,m-1} - \partial_p J_{p/2,m+1}) \right]. \end{aligned} \quad (\text{V.40})$$

where all J 's are evaluated at

$$a = \frac{\mathcal{R}^2 + \varrho^2}{2}, \quad b = \frac{\mathcal{R}^2 - \varrho^2}{2}, \quad \mathcal{R}^2 = \varrho^2 + z_c^2. \quad (\text{V.41})$$

Thus the entire m -mode puncture construction reduces to the master function

$$J_{\nu m}(a, b) = D^\nu \frac{\Gamma(\nu + 1)}{\Gamma(\nu + |m| + 1)} P_\nu^{|m|} \left(\frac{a}{D} \right), \quad D = \sqrt{a^2 - b^2}. \quad (\text{V.42})$$

VI. PROJECTIVE REDUCTION AND GENERIC-ORBIT EXTENSION OF THE PUNCTURE INTEGRALS

A. Circular-orbit master integral

The fundamental object is the Fourier coefficient

$$J_{\nu m}(a, b) := \frac{1}{2\pi} \int_{-\pi}^{\pi} (a - b \cos \psi)^\nu e^{-im\psi} d\psi.$$

For a puncture monomial R^p , one has

$$\nu = \frac{p}{2},$$

so that

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} R(\psi)^p e^{-im\psi} d\psi = J_{p/2, m}(a, b).$$

With the Condon–Shortley convention for associated Legendre functions, one convenient closed form is

$$J_{\nu m}(a, b) = (-1)^{|m|} D^\nu \frac{\Gamma(\nu + 1)}{\Gamma(\nu + |m| + 1)} P_\nu^{|m|} \left(\frac{a}{D} \right).$$

Equivalently,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} R(\psi)^p e^{-im\psi} d\psi = (-1)^{|m|} (\rho \mathcal{R})^{p/2} \frac{\Gamma(\frac{p}{2} + 1)}{\Gamma(\frac{p}{2} + |m| + 1)} P_{p/2}^{|m|} \left(\frac{\rho^2 + \frac{1}{2} z_c^2}{\rho \mathcal{R}} \right).$$

Depending on the associated-Legendre convention used by a computer algebra system, the overall factor $(-1)^{|m|}$ may be absorbed into the definition of $P_\nu^{|m|}$.

B. Reduction of sine insertions

The reduced puncture contains at most one explicit factor of

$$Z' = z_c \sin \psi.$$

Thus we need integrals of the form

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} z_c \sin \psi R(\psi)^p e^{-im\psi} d\psi.$$

These are not new master integrals because

$$\sin \psi e^{-im\psi} = \frac{1}{2i} (e^{-i(m-1)\psi} - e^{-i(m+1)\psi}).$$

Therefore

$$\mathcal{I}_{p,m}^{(1,0)} := \frac{1}{2\pi} \int_{-\pi}^{\pi} z_c \sin \psi R(\psi)^p e^{-im\psi} d\psi = \frac{z_c}{2i} [J_{p/2,m-1} - J_{p/2,m+1}].$$

Thus one power of Z' simply shifts the azimuthal mode number by $m \mapsto m \pm 1$.

More generally, any polynomial numerator in $\sin \psi$ and $\cos \psi$ can be written as a finite Fourier series:

$$\sin^r \psi \cos^s \psi = \sum_{\ell=-(r+s)}^{r+s} c_{\ell} e^{i\ell\psi}.$$

Hence

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \sin^r \psi \cos^s \psi R(\psi)^p e^{-im\psi} d\psi = \sum_{\ell} c_{\ell} J_{p/2,m-\ell}.$$

Polynomial numerator factors never introduce new master integrals; they only produce finite mode mixing.

C. Reduction of logarithmic integrals

The logarithmic integrals also follow from $J_{\nu m}$. Since

$$\frac{\partial}{\partial p} R(\psi)^p = R(\psi)^p \log R(\psi),$$

we have

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} R(\psi)^p \log R(\psi) e^{-im\psi} d\psi = \partial_p J_{p/2,m}.$$

Equivalently, writing $\nu = p/2$,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} R(\psi)^p \log R(\psi) e^{-im\psi} d\psi = \frac{1}{2} \partial_{\nu} J_{\nu m} \Big|_{\nu=p/2}.$$

For the pure logarithmic case $p = 0$, define

$$\Lambda_m := \frac{1}{2\pi} \int_{-\pi}^{\pi} \log R(\psi) e^{-im\psi} d\psi.$$

Then

$$\Lambda_m = \frac{1}{2} \partial_{\nu} J_{\nu m} \Big|_{\nu=0}.$$

There is also a direct closed form. Write

$$a - b \cos \psi = C (1 - 2\lambda \cos \psi + \lambda^2),$$

where

$$\lambda = \frac{b}{a+D} = \frac{\mathcal{R}-\rho}{\mathcal{R}+\rho}, \quad C = \frac{a+D}{2} = \frac{(\mathcal{R}+\rho)^2}{4}.$$

Then

$$R(\psi) = \frac{\mathcal{R}+\rho}{2} (1 - 2\lambda \cos \psi + \lambda^2)^{1/2}.$$

Using

$$\log(1 - 2\lambda \cos \psi + \lambda^2) = -2 \sum_{n=1}^{\infty} \frac{\lambda^n}{n} \cos(n\psi), \quad |\lambda| < 1,$$

we get

$$\Lambda_0 = \log\left(\frac{\mathcal{R}+\rho}{2}\right),$$

and for $|m| \geq 1$,

$$\Lambda_m = -\frac{1}{2|m|} \left(\frac{\mathcal{R}-\rho}{\mathcal{R}+\rho}\right)^{|m|}.$$

The logarithmic sine integral is again just a shift:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} z_c \sin \psi R(\psi)^p \log R(\psi) e^{-im\psi} d\psi = \frac{z_c}{2i} [\partial_p J_{p/2, m-1} - \partial_p J_{p/2, m+1}]. \quad (\text{VI.1})$$

For $p = 0$, this becomes

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} z_c \sin \psi \log R(\psi) e^{-im\psi} d\psi = \frac{z_c}{2i} [\Lambda_{m-1} - \Lambda_{m+1}]. \quad (\text{VI.2})$$

D. Projective representation

Now set

$$u = e^{i\psi}.$$

Then

$$d\psi = \frac{du}{iu}, \quad \cos \psi = \frac{1}{2}(u + u^{-1}), \quad e^{-im\psi} = u^{-m}.$$

Therefore

$$R(\psi)^2 = a - \frac{b}{2}(u + u^{-1}) = \frac{P(u)}{u},$$

where

$$P(u) := -\frac{b}{2}u^2 + au - \frac{b}{2}.$$

Thus

$$J_{\nu m} = \frac{1}{2\pi i} \oint_{|u|=1} u^{-m-1-\nu} P(u)^{\nu} du.$$

Equivalently, with

$$P(u) = Au^2 + Bu + C, \quad A = -\frac{b}{2}, \quad B = a, \quad C = -\frac{b}{2},$$

we have

$$J_{\nu m} = \frac{1}{2\pi i} \oint_{|u|=1} u^{-m-1-\nu} (Au^2 + Bu + C)^\nu du.$$

E. Homogenization on $\mathbb{P}^1$

Let

$$u = \frac{U}{V}. \tag{VI.3}$$

The quadratic $P(u)$ homogenizes to

$$F(U, V) = AU^2 + BUV + CV^2 = aUV - \frac{b}{2}(U^2 + V^2). \tag{VI.4}$$

The standard projective one-form on $\mathbb{P}^1$ is

$$\Omega_{\mathbb{P}^1} = U dV - V dU. \tag{VI.5}$$

Since

$$du = \frac{V dU - U dV}{V^2} = -\frac{\Omega_{\mathbb{P}^1}}{V^2}, \tag{VI.6}$$

the affine differential

$$u^{-m-1-\nu} P(u)^\nu du \tag{VI.7}$$

can be written projectively as

$$\omega_{\nu m} = -U^{-m-1-\nu} V^{m-\nu-1} F(U, V)^\nu \Omega_{\mathbb{P}^1}. \tag{VI.8}$$

Indeed, under

$$(U, V) \mapsto (\lambda U, \lambda V),$$

we have

$$F \mapsto \lambda^2 F, \quad \Omega_{\mathbb{P}^1} \mapsto \lambda^2 \Omega_{\mathbb{P}^1},$$

and the total homogeneity of $\omega_{\nu m}$ is zero. Thus

$$\boxed{\omega_{\nu m} \text{ is a well-defined projective one-form on } \mathbb{P}^1, \text{ possibly with branch structure.}}$$

For integer ν , the integrand is meromorphic on $\mathbb{P}^1$. For half-integer ν , the form is naturally defined on a double cover.

Thus the m -mode integrals are contour integrals of homogeneous projective differentials on $\mathbb{P}^1$. When $\nu \in \mathbb{Z}$, the integrand is meromorphic on $\mathbb{P}^1$. When ν is half-integer, as occurs for odd powers R^p , the integrand is naturally defined on a two-sheeted cover.

For odd powers, introduce

$$w^2 = uP(u).$$

Then

$$R(\psi)^2 = \frac{P(u)}{u}, \quad R(\psi) = \frac{w}{u}.$$

Hence odd powers of R become meromorphic differentials on the algebraic curve

$$\boxed{w^2 = uP(u)}.$$

This is the geometric meaning of the m -mode integrals: they are periods of a small family of algebraic differentials on $\mathbb{P}^1$, or on a two-sheeted cover for odd powers.

F. Exact-differential recurrence

The projective viewpoint gives algebraic recurrences. Define the more general contour integral

$$I_k^{(\nu)} := \oint u^k P(u)^\nu du, \quad P(u) = Au^2 + Bu + C. \quad (\text{VI.9})$$

The master integral is the special case

$$J_{\nu m} = \frac{1}{2\pi i} I_{-m-1-\nu}^{(\nu)}.$$

Now use the fact that the integral of an exact differential around a closed cycle vanishes:

$$0 = \oint d(u^{k+1} P(u)^{\nu+1}).$$

Expanding,

$$d(u^{k+1} P^{\nu+1}) = [(k+1)u^k P^{\nu+1} + (\nu+1)u^{k+1} P^\nu P'(u)] du.$$

Since

$$P'(u) = 2Au + B,$$

we obtain

$$0 = \oint [(k+1)u^k P^{\nu+1} + (\nu+1)u^{k+1} P^\nu (2Au + B)] du.$$

Using

$$P(u) = Au^2 + Bu + C,$$

the first term becomes

$$(k+1)u^k P^{\nu+1} = (k+1) [Au^{k+2} + Bu^{k+1} + Cu^k] P^\nu.$$

Therefore

$$0 = A(k+2\nu+3)I_{k+2}^{(\nu)} + B(k+\nu+2)I_{k+1}^{(\nu)} + C(k+1)I_k^{(\nu)}.$$

Thus

$$\boxed{A(k+2\nu+3)I_{k+2}^{(\nu)} + B(k+\nu+2)I_{k+1}^{(\nu)} + C(k+1)I_k^{(\nu)} = 0.} \quad (\text{VI.10})$$

This is the basic projective recurrence. It expresses one moment integral in terms of two neighboring ones. Because

$$A = C = -\frac{b}{2}, \quad B = a,$$

the recurrence is explicitly

$$\boxed{-\frac{b}{2}(k+2\nu+3)I_{k+2}^{(\nu)} + a(k+\nu+2)I_{k+1}^{(\nu)} - \frac{b}{2}(k+1)I_k^{(\nu)} = 0.} \quad (\text{VI.11})$$

G. Recurrence directly for $J_{\nu m}$

Since

$$J_{\nu m} = \frac{1}{2\pi i} I_{-m-1-\nu}^{(\nu)},$$

we set

$$k = -m - 1 - \nu.$$

Then

$$I_k^{(\nu)} \leftrightarrow J_{\nu m}, \quad I_{k+1}^{(\nu)} \leftrightarrow J_{\nu, m-1}, \quad I_{k+2}^{(\nu)} \leftrightarrow J_{\nu, m-2}.$$

Substituting into the exact-differential recurrence gives

$$\boxed{A(\nu - m + 2)J_{\nu, m-2} + B(1 - m)J_{\nu, m-1} - C(m + \nu)J_{\nu m} = 0.}$$

For the circular puncture case,

$$A = C = -\frac{b}{2}, \quad B = a,$$

so

$$\boxed{-\frac{b}{2}(\nu - m + 2)J_{\nu, m-2} + a(1 - m)J_{\nu, m-1} + \frac{b}{2}(m + \nu)J_{\nu m} = 0.} \quad (\text{VI.12})$$

Equivalently,

$$\boxed{J_{\nu m} = \frac{b(\nu - m + 2)J_{\nu, m-2} - 2a(1 - m)J_{\nu, m-1}}{b(m + \nu)}}.$$

This form is useful for generating all m -modes from a small number of seed values, subject to the usual care near exceptional indices where the denominator vanishes.

H. Logarithmic recurrences

Define the logarithmic moment

$$K_k^{(\nu)} := \partial_\nu I_k^{(\nu)} = \oint u^k P(u)^\nu \log P(u) du.$$

Differentiating the recurrence

$$A(k + 2\nu + 3)I_{k+2}^{(\nu)} + B(k + \nu + 2)I_{k+1}^{(\nu)} + C(k + 1)I_k^{(\nu)} = 0$$

with respect to ν gives

$$\boxed{A(k + 2\nu + 3)K_{k+2}^{(\nu)} + B(k + \nu + 2)K_{k+1}^{(\nu)} + C(k + 1)K_k^{(\nu)} + 2AI_{k+2}^{(\nu)} + BI_{k+1}^{(\nu)} = 0.} \quad (\text{VI.13})$$

This recurrence reduces logarithmic integrals to the same seed data plus the already-computed algebraic moments.

For the physical logarithm $\log R$, note that

$$R^{2\nu} = u^{-\nu} P(u)^\nu.$$

Therefore differentiating $J_{\nu m}$ gives

$$\partial_\nu J_{\nu m} = \frac{1}{2\pi i} \oint u^{-m-1-\nu} P(u)^\nu [\log P(u) - \log u] du.$$

Since

$$R^p \log R = \frac{1}{2} \partial_\nu R^{2\nu} \Big|_{\nu=p/2},$$

the physical logarithmic integral is

$$\boxed{\frac{1}{2\pi} \int_{-\pi}^{\pi} R(\psi)^p \log R(\psi) e^{-im\psi} d\psi = \frac{1}{2} \partial_\nu J_{\nu m} \Big|_{\nu=p/2}.} \quad (\text{VI.14})$$

VII. EXTENSION TO NON-CIRCULAR ORBITS

For a circular orbit, one can often choose a comoving, rotating frame in which the local field is independent of the comoving time coordinate T . This is ultimately a consequence of the helical symmetry generated by

$$k^\mu = t^\mu + \Omega \phi^\mu. \quad (\text{VII.1})$$

In a frame adapted to the circular motion, the condition

$$\left(\frac{\partial}{\partial T} \right)^\mu = u^\mu \quad (\text{VII.2})$$

turns comoving stationarity into

$$\partial_T F_{A_1 \dots A_p} = 0. \quad (\text{VII.3})$$

The field then admits a purely spatial STF or tensor-spherical-harmonic expansion.

For a non-circular orbit, there is in general no helical Killing vector. The worldline

$$x^\mu = z^\mu(T) \quad (\text{VII.4})$$

has time-dependent position, velocity, and tetrad orientation. We may still introduce a local comoving frame by choosing an orthonormal tetrad

$$e_0^\mu(T) = u^\mu(T), \quad g_{\mu\nu} e_i^\mu e_j^\nu = \delta_{ij}, \quad (\text{VII.5})$$

and defining local coordinates by

$$x^\mu = \exp_{z(T)} \left(X^i e_i^\mu(T) \right). \quad (\text{VII.6})$$

The adapted time coordinate satisfies, at least on the worldline,

$$\left(\frac{\partial}{\partial T} \right)_\gamma^\mu = u^\mu. \quad (\text{VII.7})$$

One may extend this off the worldline by choosing a timelike congruence whose flow lines carry the spatial labels X^i . In that case,

$$u^\mu \nabla_\mu X^i = 0, \quad u^\mu \nabla_\mu T = 1. \quad (\text{VII.8})$$

However, unlike the circular case, the local field is not generally stationary:

$$\mathcal{L}_u F_{\mu_1 \dots \mu_p} \neq 0. \quad (\text{VII.9})$$

Therefore the tetrad components obey

$$F_{A_1 \dots A_p} = F_{A_1 \dots A_p}(T, X^i). \quad (\text{VII.10})$$

Introduce

$$R = \sqrt{\delta_{ij} X^i X^j}, \quad n^i = \frac{X^i}{R}. \quad (\text{VII.11})$$

At each instant T , we expand the angular dependence in STF tensors:

$$\boxed{F_{A_1 \dots A_p}(T, R, n^i) = \sum_{\ell=0}^{\infty} F_{A_1 \dots A_p L}(T, R) n^{\langle L \rangle}.} \quad (\text{VII.12})$$

The tensorial character of the field is carried by the tetrad indices

$$A_1, \dots, A_p, \quad (\text{VII.13})$$

while the angular multipole structure is carried by the STF multi-index

$$L = i_1 \dots i_\ell. \quad (\text{VII.14})$$

The only difference from the circular case is that the STF multipole coefficients are now functions of time:

$$F_{A_1 \dots A_p L} = F_{A_1 \dots A_p L}(T, R). \quad (\text{VII.15})$$

For a bound orbit, the motion is quasi-periodic. Instead of using T directly, introduce orbital phases

$$q_a(T), \quad a \in \{r, \theta, \phi\},$$

with fundamental frequencies

$$\frac{dq_a}{dT} = \Omega_a.$$

For a bound Kerr geodesic, one often uses the radial and polar phases

$$q_r, \quad q_\theta,$$

together with the azimuthal phase. Then the local multipole coefficients may be expanded as Fourier series on the orbital torus:

$$\boxed{F_{A_1 \dots A_p L}(T, R) = \sum_{\mathbf{k} \in \mathbb{Z}^N} F_{A_1 \dots A_p L, \mathbf{k}}(R) e^{-i\mathbf{k} \cdot \mathbf{q}(T)}.} \quad (\text{VII.16})$$

For a generic bound Kerr orbit, one may write schematically

$$\mathbf{k} \cdot \mathbf{q} = k_r q_r + k_\theta q_\theta + m q_\phi.$$

Thus

$$F_{A_1 \dots A_p}(T, R, n^i) = \sum_{\ell, \mathbf{k}} F_{A_1 \dots A_p L, \mathbf{k}}(R) n^{\langle L \rangle} e^{-i \mathbf{k} \cdot \mathbf{q}(T)}.$$

$$\text{circular orbit} \quad : F_{A_1 \dots A_p}(R, n^i) = \sum_{\ell} F_{A_1 \dots A_p L}(R) n^{\langle L \rangle},$$

$$\text{non-circular orbit} : F_{A_1 \dots A_p}(q_a; R, n^i) = \sum_{\ell, \mathbf{k}} F_{A_1 \dots A_p L, \mathbf{k}}(R) n^{\langle L \rangle} e^{-i \mathbf{k} \cdot \mathbf{q}}.$$

A. Generic bound-orbit extension

For a circular equatorial orbit, the orbit data are constants:

$$r_0, \quad \Omega, \quad z_c, \quad \mathcal{B}, \quad \mathcal{D}.$$

For a generic bound Kerr orbit, these become functions of the orbital phases. Let

$$q := (q_r, q_\theta)$$

denote the radial and polar phases. Then

$$r_p = r_p(q_r), \quad \theta_p = \theta_p(q_\theta),$$

and the oscillatory pieces of t_p and ϕ_p are functions on the two-torus:

$$\Delta t_p = \Delta t_p(q_r, q_\theta), \quad \Delta \phi_p = \Delta \phi_p(q_r, q_\theta).$$

The local co-moving coordinates should be promoted to phase-dependent coordinates,

$$X = X(x^\mu; q), \quad Y = Y(x^\mu; q), \quad Z = Z(x^\mu; q), \quad (\text{VII.17})$$

with the particle at

$$X = Y = Z = 0. \quad (\text{VII.18})$$

Schematically,

$$X \sim \frac{r - r_p(q_r)}{\mathcal{B}(q)},$$

$Y \sim \text{local polar displacement from } \theta_p(q_\theta),$

and

$$Z \sim z_c(q) \sin \left(\frac{\phi - \phi_p(q)}{2} \right).$$

The puncture field then has the same local structure,

$$h_{ab}^P = \sum_{i,j,k,p,q_\ell} c_{ijkpq_\ell}^{ab}(q) X^i Y^j Z^k R^p \log^{q_\ell} R,$$

but now the coefficients are functions on the orbital torus:

$$c_{ijkpq_\ell}^{ab} = c_{ijkpq_\ell}^{ab}(q_r, q_\theta).$$

B. Generic local m -mode integral

For a generic orbit, the squared local distance as a function of the azimuthal separation need not be exactly $a - b \cos \psi$. The most general quadratic dependence generated by one azimuthal angle has the form

$$R_q(\psi)^2 = a(q) - b(q) \cos \psi - c(q) \sin \psi,$$

where

$$\psi := \phi - \phi_p(q).$$

Define

$$\beta(q) := \sqrt{b(q)^2 + c(q)^2},$$

and choose $\delta(q)$ such that

$$b(q) = \beta(q) \cos \delta(q), \quad c(q) = \beta(q) \sin \delta(q).$$

Then

$$b(q) \cos \psi + c(q) \sin \psi = \beta(q) \cos(\psi - \delta(q)).$$

Hence

$$R_q(\psi)^2 = a(q) - \beta(q) \cos(\psi - \delta(q)).$$

The generic local master integral is therefore

$$J_{\nu m}(q) := \frac{1}{2\pi} \int_{-\pi}^{\pi} [a(q) - b(q) \cos \psi - c(q) \sin \psi]^\nu e^{-im\psi} d\psi.$$

Shifting

$$\chi := \psi - \delta(q)$$

gives

$$J_{\nu m}(q) = e^{-im\delta(q)} J_{\nu m}(a(q), \beta(q)).$$

Thus the same associated-Legendre formula applies:

$$J_{\nu m}(q) = e^{-im\delta(q)} (-1)^{|m|} D(q)^\nu \frac{\Gamma(\nu + 1)}{\Gamma(\nu + |m| + 1)} P_\nu^{|m|} \left(\frac{a(q)}{D(q)} \right), \quad (\text{VII.19})$$

where

$$D(q) := \sqrt{a(q)^2 - \beta(q)^2}. \quad (\text{VII.20})$$

The projective representation also carries over. With $u = e^{i\psi}$,

$$R_q(\psi)^2 = a(q) - \frac{1}{2}(b(q) - ic(q))u - \frac{1}{2}(b(q) + ic(q))u^{-1}.$$

Thus

$$R_q(\psi)^2 = \frac{P_q(u)}{u},$$

with

$$P_q(u) = -\frac{1}{2}(b(q) - ic(q))u^2 + a(q)u - \frac{1}{2}(b(q) + ic(q)).$$

Hence

$$J_{\nu m}(q) = \frac{1}{2\pi i} \oint_{|u|=1} u^{-m-1-\nu} P_q(u)^\nu du. \quad (\text{VII.21})$$

The exact-differential recurrence remains valid, but with phase-dependent coefficients:

$$P_q(u) = A(q)u^2 + B(q)u + C(q),$$

where

$$A(q) = -\frac{1}{2}(b(q) - ic(q)), \quad B(q) = a(q), \quad C(q) = -\frac{1}{2}(b(q) + ic(q)).$$

Therefore

$$A(q)(k + 2\nu + 3)I_{k+2}^{(\nu)}(q) + B(q)(k + \nu + 2)I_{k+1}^{(\nu)}(q) + C(q)(k + 1)I_k^{(\nu)}(q) = 0. \quad (\text{VII.22})$$

C. From local m -modes to frequency modes

For a circular orbit, helical symmetry gives

$$\omega = m\Omega_\phi.$$

For a generic bound Kerr orbit, the frequency spectrum is

$$\boxed{\omega_{mkn} = m\Omega_\phi + k\Omega_\theta + n\Omega_r.} \quad (\text{VII.23})$$

The generic construction has two layers.

First, compute the local azimuthal m -mode at fixed orbital phase:

$$\tilde{h}_{ab}^{P,m}(q; r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} h_{ab}^P(q; r, \theta, \psi) e^{-im\psi} d\psi.$$

This is where the local integrals

$$J_{\nu m}(q), \quad \partial_\nu J_{\nu m}(q), \quad J_{\nu, m\pm 1}(q), \quad \partial_\nu J_{\nu, m\pm 1}(q)$$

enter.

Second, Fourier transform over the orbital torus. With the mode convention

$$e^{-i\omega t + im\phi},$$

one convenient schematic formula is

$$\boxed{h_{ab; mkn}^P(r, \theta) = \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \tilde{h}_{ab}^{P,m}(q_r, q_\theta; r, \theta) e^{i\omega_{mkn}\Delta t(q_r, q_\theta)} e^{-im\Delta\phi(q_r, q_\theta)} e^{-i(nq_r + kq_\theta)} dq_r dq_\theta.} \quad (\text{VII.24})$$

Here Δt and $\Delta\phi$ are the oscillatory pieces of the orbital functions, after removing the secular terms. The precise signs depend on the Fourier convention, but the structure is universal:

$$\boxed{\text{generic frequency mode} = \text{torus Fourier transform of the local } m\text{-mode.}}$$

The circular limit is obtained when all coefficients are constant on the torus and the oscillatory orbital phases vanish. Then the torus integral collapses, and one recovers the single helical frequency

$$\omega = m\Omega_\phi.$$

D. Practical reduction algorithm

The resulting workflow is:

1. Write every puncture monomial in the reduced form

$$X^i Y^j (Z')^k R^p \log^q R, \quad k, q \in \{0, 1\}.$$

2. Replace each monomial by the appropriate master integral:

$$R^p \mapsto J_{p/2, m},$$

$$Z' R^p \mapsto \frac{z_c}{2i} (J_{p/2, m-1} - J_{p/2, m+1}),$$

$$R^p \log R \mapsto \partial_p J_{p/2, m} = \frac{1}{2} \partial_\nu J_{\nu m} \Big|_{\nu=p/2},$$

and

$$Z' R^p \log R \mapsto \frac{z_c}{2i} (\partial_p J_{p/2, m-1} - \partial_p J_{p/2, m+1}).$$

3. Evaluate $J_{\nu m}$ either by the associated-Legendre formula

$$J_{\nu m} = (-1)^{|m|} D^\nu \frac{\Gamma(\nu + 1)}{\Gamma(\nu + |m| + 1)} P_\nu^{|m|} \left(\frac{a}{D} \right),$$

or by the projective recurrence

$$A(k + 2\nu + 3) I_{k+2}^{(\nu)} + B(k + \nu + 2) I_{k+1}^{(\nu)} + C(k + 1) I_k^{(\nu)} = 0.$$

4. For generic orbits, promote

$$a, b, c, z_c, \rho, \mathcal{R}$$

to functions of the orbital torus variables q_r, q_θ , reduce

$$a - b \cos \psi - c \sin \psi$$

to

$$a - \beta \cos(\psi - \delta),$$

and use

$$J_{\nu m}(q) = e^{-im\delta(q)} J_{\nu m}(a(q), \beta(q)).$$

5. Finally Fourier transform the resulting local m -mode over the orbital torus to obtain the (m, k, n) frequency modes.

1. STF tensors, helicity basis, and E/B vector and tensor harmonics

We want to relate three closely connected descriptions of angular multipoles on the sphere:

STF tensors $\longleftrightarrow$ spin-weighted sph. harmonics $\longleftrightarrow$ E/B vector & tensor harmonics.

The main idea is simple. An STF tensor carries the (ℓ, m) multipole. Its helicity projections give the spin-weighted spherical harmonics ${}_sY_{\ell m}$. The even- and odd-parity vector and tensor harmonics are then obtained by taking parity eigenstate combinations of the two helicities $s = \pm 1$ or $s = \pm 2$.

We work on the unit sphere with metric Ω_{AB} , covariant derivative D_A , and volume form ϵ_{AB} . Introduce the complex helicity dyad

$$m^A = \frac{1}{\sqrt{2}} (e_\theta^A + ie_\phi^A), \quad \bar{m}^A = \frac{1}{\sqrt{2}} (e_\theta^A - ie_\phi^A), \quad (.25)$$

so that

$$\Omega_{AB} = m_A \bar{m}_B + \bar{m}_A m_B, \quad \epsilon_A{}^B m_B = im_A, \quad \epsilon_A{}^B \bar{m}_B = -i\bar{m}_A. \quad (.26)$$

A scalar spherical harmonic may be represented by a constant STF tensor $\mathcal{Y}_L^{\ell m}$, with multi-index $L = i_1 \cdots i_\ell$, as

$$Y_{\ell m}(\hat{n}) = \mathcal{Y}_L^{\ell m} n^{\langle L \rangle}. \quad (.27)$$

Here n^i is the unit radial vector and angular brackets denote the symmetric tracefree projection.

The corresponding spin-weighted harmonics are obtained by replacing some of the radial vectors n^i by helicity vectors m^i or $\bar{m}^i$. With the conventions above,

$${}_s Y_{\ell m} = (-1)^s C_{\ell s} \mathcal{Y}_{i_1 \dots i_\ell}^{\ell m} m^{i_1} \cdots m^{i_s} n^{i_{s+1}} \cdots n^{i_\ell}, \quad (.28)$$

and

$${}_{-s} Y_{\ell m} = C_{\ell s} \mathcal{Y}_{i_1 \dots i_\ell}^{\ell m} \bar{m}^{i_1} \cdots \bar{m}^{i_s} n^{i_{s+1}} \cdots n^{i_\ell}, \quad (.29)$$

where

$$C_{\ell s} = 2^{s/2} \frac{\ell!}{(\ell-s)!} \sqrt{\frac{(\ell-s)!}{(\ell+s)!}}. \quad (.30)$$

In particular,

$${}_1 Y_{\ell m} = -\sqrt{\frac{2\ell}{\ell+1}} \mathcal{Y}_{i_{L-1}}^{\ell m} m^i n^{L-1}, \quad {}_{-1} Y_{\ell m} = +\sqrt{\frac{2\ell}{\ell+1}} \mathcal{Y}_{i_{L-1}}^{\ell m} \bar{m}^i n^{L-1}, \quad (.31)$$

and

$${}_2 Y_{\ell m} = 2\sqrt{\frac{\ell(\ell-1)}{(\ell+1)(\ell+2)}} \mathcal{Y}_{ijL-2}^{\ell m} m^i m^j n^{L-2}. \quad (.32)$$

a. Vector harmonics

The even-parity and odd-parity vector harmonics are defined by

$$Y_A^{E,\ell m} = \frac{1}{\sqrt{\ell(\ell+1)}} D_A Y_{\ell m}, \quad (.33)$$

and

$$Y_A^{B,\ell m} = \frac{1}{\sqrt{\ell(\ell+1)}} \epsilon_A^B D_B Y_{\ell m}. \quad (.34)$$

The gradient of a scalar harmonic decomposes in the helicity basis as

$$D_A Y_{\ell m} = \sqrt{\frac{\ell(\ell+1)}{2}} (-{}_1Y_{\ell m} m_A - {}_1Y_{\ell m} \bar{m}_A). \quad (.35)$$

Therefore

$$Y_A^{E,\ell m} = \frac{1}{\sqrt{2}} (-{}_1Y_{\ell m} m_A - {}_1Y_{\ell m} \bar{m}_A), \quad (.36)$$

while applying ϵ_A^B gives

$$Y_A^{B,\ell m} = \frac{i}{\sqrt{2}} (-{}_1Y_{\ell m} m_A + {}_1Y_{\ell m} \bar{m}_A). \quad (.37)$$

Equivalently, their helicity components are

$$m^A Y_A^{E,\ell m} = -\frac{1}{\sqrt{2}} {}_1Y_{\ell m}, \quad \bar{m}^A Y_A^{E,\ell m} = +\frac{1}{\sqrt{2}} {}_{-1}Y_{\ell m}, \quad (.38)$$

and

$$m^A Y_A^{B,\ell m} = +\frac{i}{\sqrt{2}} {}_1Y_{\ell m}, \quad \bar{m}^A Y_A^{B,\ell m} = +\frac{i}{\sqrt{2}} {}_{-1}Y_{\ell m}. \quad (.39)$$

The important helicity combinations are therefore

$$Y_A^{E,\ell m} + iY_A^{B,\ell m} = -\sqrt{2} {}_1Y_{\ell m} \bar{m}_A, \quad (.40)$$

and

$$Y_A^{E,\ell m} - iY_A^{B,\ell m} = +\sqrt{2} {}_{-1}Y_{\ell m} m_A. \quad (.41)$$

Thus $Y_A^{E,\ell m}$ and $Y_A^{B,\ell m}$ are parity eigenstates, whereas $Y_A^{E,\ell m} \pm iY_A^{B,\ell m}$ are helicity eigenstates.

b. Tensor harmonics

Now consider symmetric tracefree rank-two tensor harmonics on the sphere. Define the even-parity tensor harmonic by

$$Y_{AB}^{E,\ell m} = N_\ell \left(D_A D_B Y_{\ell m} - \frac{1}{2} \Omega_{AB} D^2 Y_{\ell m} \right), \quad (.42)$$

and the odd-parity tensor harmonic by

$$Y_{AB}^{B,\ell m} = \frac{N_\ell}{2} (\epsilon_A^C D_B D_C Y_{\ell m} + \epsilon_B^C D_A D_C Y_{\ell m}), \quad (.43)$$

where

$$N_\ell = \sqrt{\frac{2(\ell-2)!}{(\ell+2)!}}. \quad (.44)$$

With this normalization, the helicity decomposition is

$$Y_{AB}^{E,\ell m} = \frac{1}{\sqrt{2}} ({}_2Y_{\ell m} \bar{m}_A \bar{m}_B + {}_{-2}Y_{\ell m} m_A m_B), \quad (.45)$$

and

$$Y_{AB}^{B,\ell m} = \frac{i}{\sqrt{2}} ({}_2Y_{\ell m} m_A m_B - {}_2Y_{\ell m} \bar{m}_A \bar{m}_B). \quad (.46)$$

Equivalently,

$$m^A m^B Y_{AB}^{E,\ell m} = \frac{1}{\sqrt{2}} {}_2Y_{\ell m}, \quad \bar{m}^A \bar{m}^B Y_{AB}^{E,\ell m} = \frac{1}{\sqrt{2}} {}_{-2}Y_{\ell m}, \quad (.47)$$

and

$$m^A m^B Y_{AB}^{B,\ell m} = -\frac{i}{\sqrt{2}} {}_2Y_{\ell m}, \quad \bar{m}^A \bar{m}^B Y_{AB}^{B,\ell m} = +\frac{i}{\sqrt{2}} {}_{-2}Y_{\ell m}. \quad (.48)$$

The pure-helicity tensor combinations are therefore

$$Y_{AB}^{E,\ell m} + iY_{AB}^{B,\ell m} = \sqrt{2} {}_2Y_{\ell m} \bar{m}_A \bar{m}_B, \quad (.49)$$

and

$$Y_{AB}^{E,\ell m} - iY_{AB}^{B,\ell m} = \sqrt{2} {}_{-2}Y_{\ell m} m_A m_B. \quad (.50)$$

c. Parity and helicity

The E - and B -type harmonics are parity eigenstates. Under $\hat{n} \mapsto -\hat{n}$,

$$Y^{E,\ell m} \sim (-1)^\ell, \quad Y^{B,\ell m} \sim (-1)^{\ell+1}. \quad (.51)$$

However, the combinations

$$E + iB, \quad E - iB, \quad (.52)$$

are not parity eigenstates. Instead, they are helicity eigenstates. For vector harmonics these have spin weights $+1$ and -1 , and for tensor harmonics they have spin weights $+2$ and -2 .